Project Final Report

Project Name	Automated Employee Onboarding & Offboarding System
Platform	ServiceNow
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Team ID	SkillWallet — ServiceNow Project
Date	15 October 2026

1. INTRODUCTION

1.1 Project Overview

The "Automated Employee Onboarding & Offboarding System" is a ServiceNow-based application designed to streamline the entire employee lifecycle. It replaces manual, siloed HR, IT, Facilities, and Security tasks with a unified, automated workflow built using ServiceNow Flow Designer.

1.2 Purpose

The purpose of this project is to reduce manual coordination, eliminate incomplete requests, ensure role-based data security, and provide a fully auditable, end-to-end lifecycle tracking system for new hires and exiting employees.

2. IDEATION PHASE

2.1 Problem Statement

How might we automate the employee onboarding and offboarding process in ServiceNow to eliminate manual coordination between HR, IT, Facilities, and Security—reducing onboarding time from days to hours while maintaining secure role-based access and complete audit visibility?

2.2 Empathy Map Canvas (HR Manager Persona)

- **Hear:** "Are we ready for Monday?", "IT hasn't set up the laptop", "Who owns this task?"
- **Think & Feel:** Overwhelmed by chasing people, anxious about missing critical steps.
- **See:** Endless email threads, spreadsheets, frustrated new hires.
- **Say & Do:** Sends manual emails, follows up constantly, apologizes for delays.
- **Pain:** Manual coordination, missing request data, no status visibility.
- **Gain:** Fast onboarding, unified process, zero missed tasks, complete audit trail.

2.3 Brainstorming

- **Ideas Generated:** 40+ across 8 clusters (Intake, Approvals, Tasks, Data, Security, Notifications, SLAs, Testing).
- **Top Priority (Must-Build):** Catalog item, custom lifecycle table, Flow Designer automation (15 actions), manager approval routing, auto-task creation (IT, Facilities, Security), and role-based ACLs.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- **Request:** HR fills a single portal form (dynamic fields).
- **Approve:** Manager gets an email and approves via ServiceNow.
- **Fulfill:** 3 tasks auto-created. IT, Facilities, and Security complete them.

- **Complete:** Flow updates Lifecycle status to "Completed" and closes RITM.
- **Extend:** Audit trail is saved for compliance; HR repeats for the next hire.

3.2 Solution Requirement

- **Functional:** 12 Requirements (Form submission, manager approval, 3 task auto-creation, status updates, ACLs, notifications).
- **Non-Functional:** 8 Requirements (Usability, Security, Reliability, Performance, Availability, Scalability, Maintainability, Auditability).

3.3 Data Flow Diagram (DFD)

- **Level 0:** HR submits form → ServiceNow Platform processes → Manager Approves → Tasks created for IT/Facilities/Security → Lifecycle table updated.
- **Data Stores:** u_employee_lifecycle (custom table), sc_request, sc_req_item, sc_task, sysapproval_approver.

3.4 Technology Stack

- **Frontend:** Service Portal (AngularJS based).
- **Backend / App Logic:** ServiceNow Flow Designer (Low-code).
- **Database:** ServiceNow Cloud DB (MySQL variant).
- **Security:** ServiceNow ACLs (Access Control Lists) + Roles.
- **Integrations (Future):** Integration Hub (AD, Slack, SAP).

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution
Manual emails causing delays.	Flow Designer auto-routing.
Missing data.	Mandatory catalog fields.
No visibility.	Custom Lifecycle table with real-time status.

4.2 Proposed Solution

A 100% low-code ServiceNow application. It uses a 13-variable catalog form to capture data. Upon submission, a Flow Designer workflow triggers, requests dynamic approval from the manager, and upon approval, creates 3 departmental tasks. Once all tasks close, the system automatically updates the lifecycle record, closes the RITM, and sends notifications.

4.3 Solution Architecture

- **Presentation Layer:** Service Portal (/sp), Agent UI, Dashboards.
- **Application Layer:** Service Catalog, Flow Designer (15 actions), Approval Engine, Task Engine.
- **Data Layer:** Custom table u_employee_lifecycle, sc_task, sys_email.
- **Security Layer:** ACLs, Roles, Groups, Audit Logs.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- **Methodology:** Agile (Scrum).
- **Sprints:** 4 Sprints.
- **Total User Stories:** 25.
- **Total Story Points:** 70.

- **Sprint Duration:** 6 days per sprint.
- **Total Duration:** 24 days.
- **Velocity:** ~17.5 story points per sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

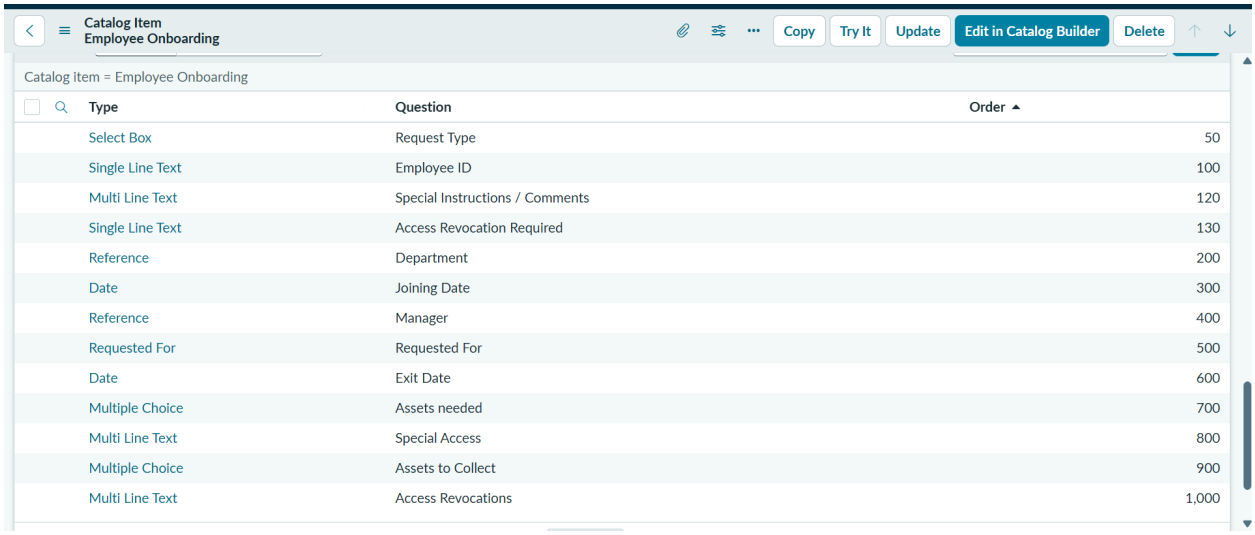
6.1 Performance Testing

- **Flow Execution Runtime:** 1,347 ms (approx. 1.35 seconds) per trigger.
- **Load Capacity:** Successfully tested concurrent requests without performance degradation.
- **UAT Result:** 16/16 test cases passed (100% success rate).
- **Bugs Fixed:** 4 (Submitted Request mapping, Email recipient, Choice value cache, GitHub nesting).

7. RESULTS

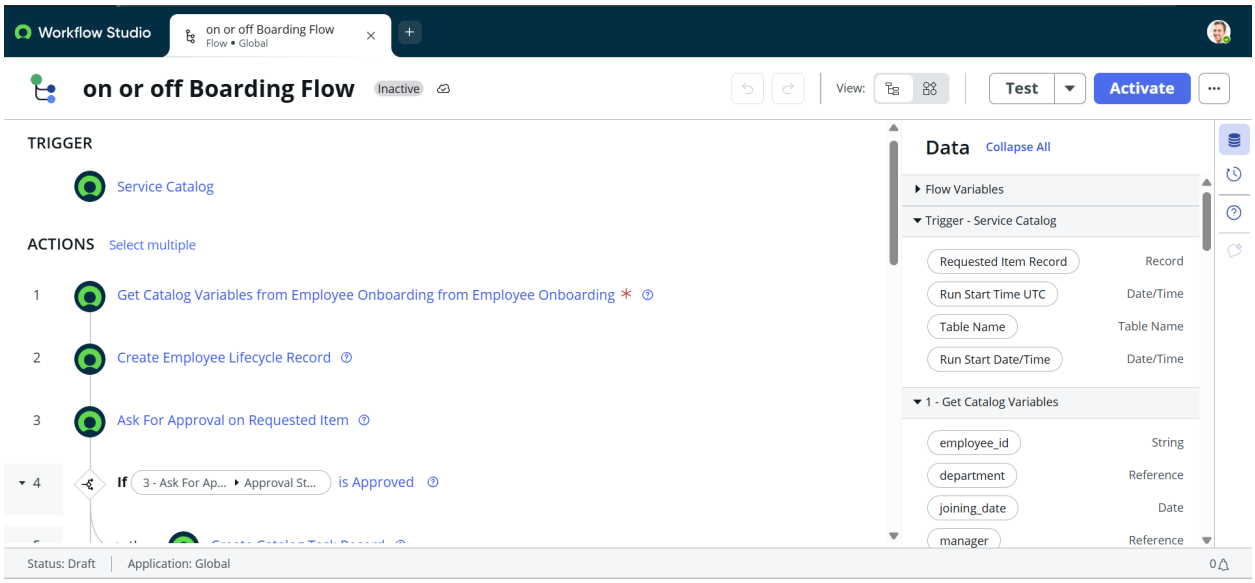
7.1 Output Screenshots

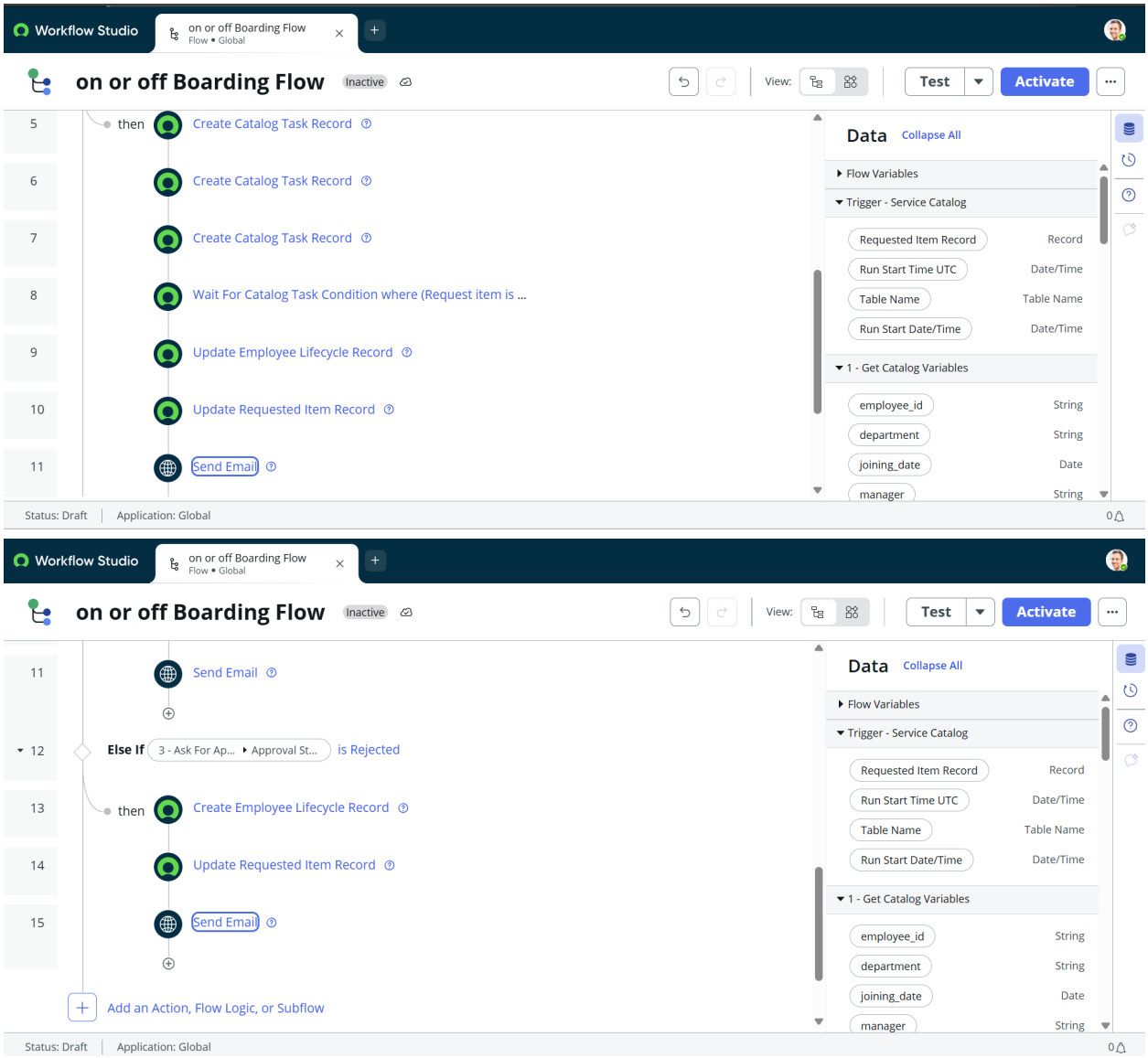
Screenshot 1: Service Portal Catalog Item (13 variables).



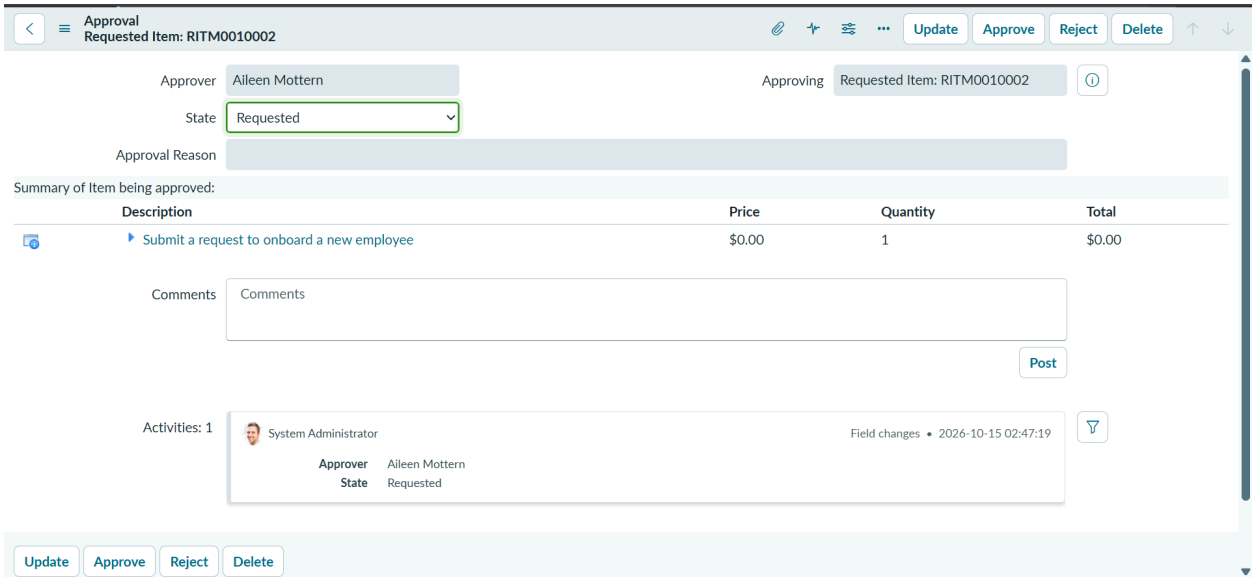
Type	Question	Order
Select Box	Request Type	50
Single Line Text	Employee ID	100
Multi Line Text	Special Instructions / Comments	120
Single Line Text	Access Revocation Required	130
Reference	Department	200
Date	Joining Date	300
Reference	Manager	400
Requested For	Requested For	500
Date	Exit Date	600
Multiple Choice	Assets needed	700
Multi Line Text	Special Access	800
Multiple Choice	Assets to Collect	900
Multi Line Text	Access Revocations	1,000

Screenshot 2: Flow Designer Active Canvas (15 actions).

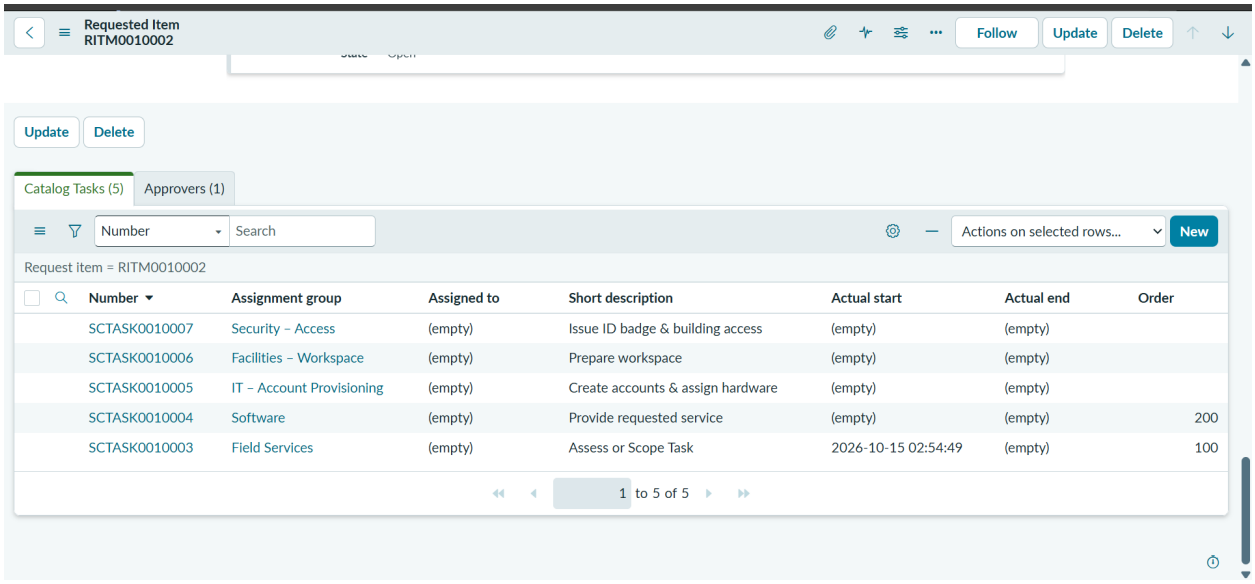




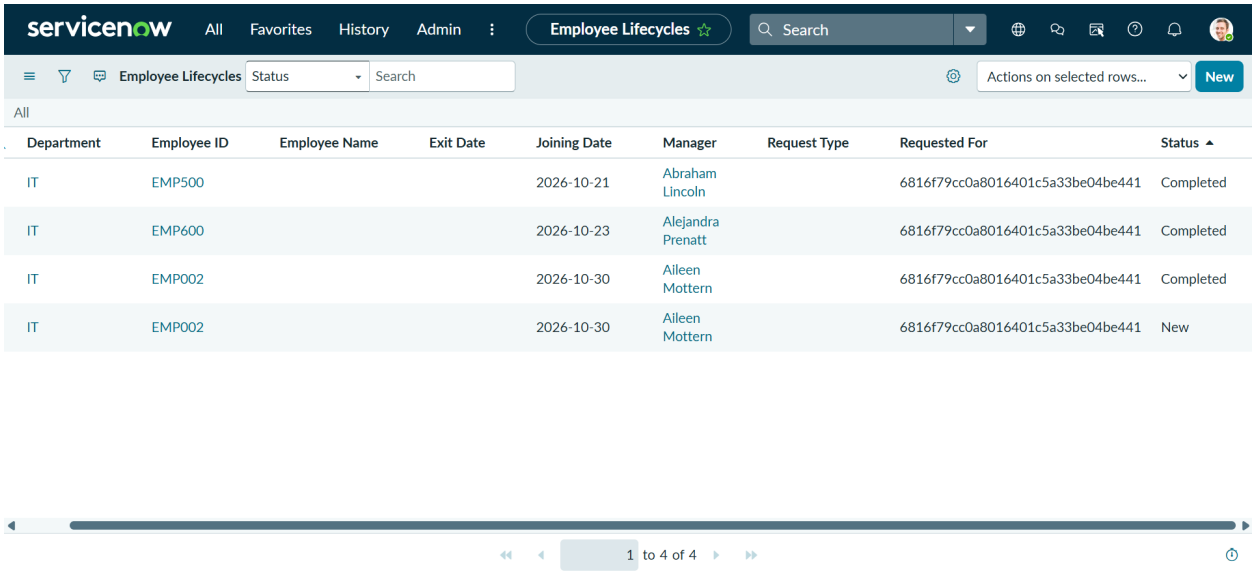
Screenshot 3: Manager Approval Record.



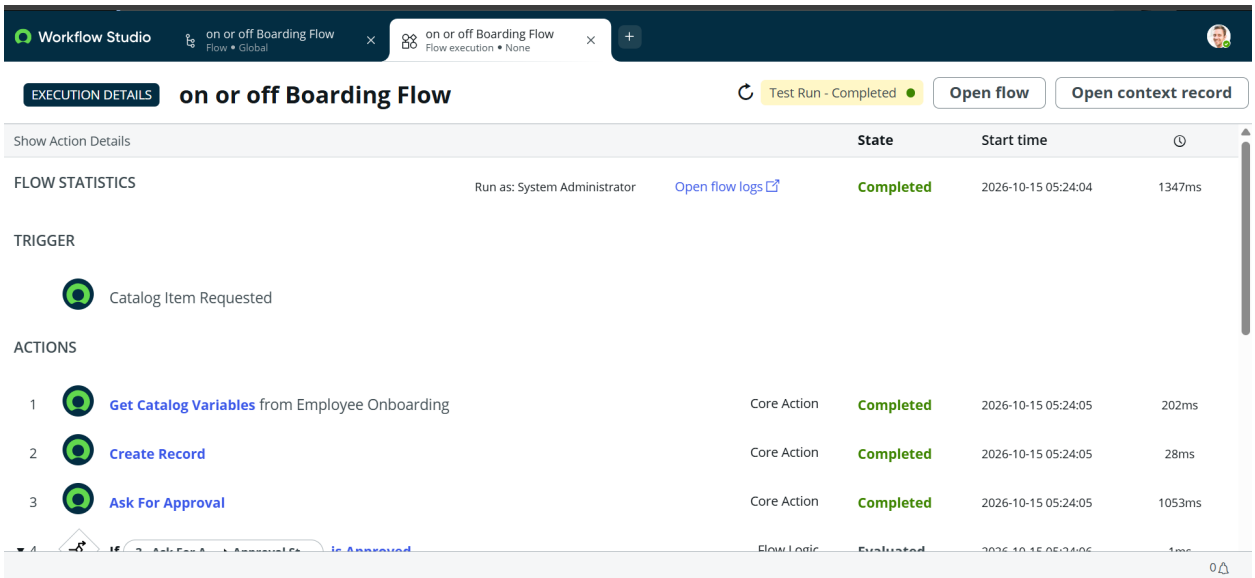
Screenshot 4: 3 Auto-generated Catalog Tasks (IT, Facilities, Security).



Screenshot 5: Employee Lifecycle Record showing Status = "Completed".



Screenshot 6: RITM showing 3 green checkmarks (Completed).



8. ADVANTAGES & DISADVANTAGES

Advantages

- **100% Low-Code:** Zero custom scripts (no Business Rules or Script Includes), making it highly maintainable.
- **Dynamic Approvals:** Routes to the correct manager automatically.
- **Role-Based Security:** ACLs ensure data confidentiality.
- **Audit Readiness:** Complete history of every approval and task.

Disadvantages

- **Platform Dependency:** Heavily reliant on the ServiceNow ecosystem.
- **Basic Native Reporting:** Without advanced Performance Analytics, dashboards are relatively basic.
- **No AD Integration (Yet):** IT still has to manually create the AD account (planned for future scope).

9. CONCLUSION

The Automated Employee Onboarding & Offboarding project successfully modernizes HR operations. By utilizing ServiceNow Flow Designer, the system eliminates manual email coordination, ensures data accuracy via mandatory catalog fields, and enforces security through role-based ACLs. End-to-end testing confirmed that requests flow seamlessly from submission to completion, proving the solution is ready for production deployment.

10. FUTURE SCOPE

- **Active Directory Integration:** Auto-create user accounts via Integration Hub upon task approval.
- **Performance Analytics Dashboards:** Build advanced reporting for SLA compliance and request volume.
- **Expanded HR Services:** Add Leave Requests, Transfer Requests, and Exit Interviews using the same workflow pattern.
- **Virtual Agent Chatbot:** Allow employees to initiate onboarding requests via MS Teams or Slack.

11. APPENDIX

- **Source Code / Update Set:** Available in the update-sets/ folder of the GitHub repo.
- **Dataset Link:** ServiceNow Instance Database (u_employee_lifecycle table).
- **GitHub Repository:** https://github.com/UmaMaheshwariKandula/servicenow-onboarding-offboarding
- **Project Demo Video:** https://drive.google.com/file/d/15bFoSLjFWv8efrGt3p6yLLCJCoEJXH1O/view?usp=sharing